

Referee report: *Holonomy and prunability as the two edges of one spectral sequence*

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Title of reviewed paper: *Holonomy and prunability as the two edges of one spectral sequence* (H²-cluster; 10pp)
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Page numbers refer to the 10-page PDF whose front matter reads “Corresponds to scotmacbeth/work-in-progress content-commit ddb7c0”. Finite computations below are in `work-in-progress/reviews/scripts/2026-09-25-h2cluster-checks.py` (bar complex over $\mathbb{F}_2$). I do not have Ferri’s paper or your Lean file locally, so I take Ferri’s Table 3 data (the loop sets) as you report them and do not re-audit the Lean kernel; nothing below depends on either.

Summary of the claims

- **Theorem 1 / 4 / 10** (pp. 2, 5, 8): for a connected groupoid G with vertex group π , normal abelian $D \trianglelefteq \pi$, $\Gamma = \pi/D$: $H_{cat}^2(G; A) \cong H^2(\pi; M)$ is the abutment of the LHS spectral sequence of $1 \rightarrow D \rightarrow \pi \rightarrow \Gamma \rightarrow 1$; the base edge $E_2^{2,0}$ is the home of the holonomy $[\omega]$; the fibre edge $E_2^{0,2}$ is the home of Ferri prunability; the transgression is $d_2(\varphi) = \varphi_*[\omega]$; and “the direct sum (1) holds if and only if $[\omega] = 0$ ”.
- **Theorem 2 / §7** (pp. 3, 7–8): on Ferri’s $K_{2,3}$ the base edge vanishes and the fibre edge carries a nonzero class $[\pi] \in H^2(\mathbb{Z}/2; \mathbb{Z}/2)$, namely the class of $0 \rightarrow \{0, 2\} \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$; a Klein-four control “kills both edges”; hence “the fibre edge is the genuine detector of prunability”.
- **Your question (email, 18 Sep)**: is “prunability = nonsplitting of $0 \rightarrow A_2 \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$ on the fibre edge” the right skew-brace-side statement, or does the dynamical non-group star break the identification?

Verdicts on the questions asked

Q1. The spectral-sequence setup. The group-cohomological core is right; the two “homes” are not.

- $H_{cat}^2(G; A) \cong H^2(\pi; M)$ for connected G , and the LHS sequence of $1 \rightarrow D \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ abutting to it: **fine** (standard; equivalence invariance of groupoid cohomology).
- Transgression = pushforward of the extension class: **fine when D acts trivially on M** , up to sign. Redoing your cocycle computation with $g = d s(a)$, $h = e s(b)$, $gh = d({}^a e) \omega(a, b) s(ab)$ and $\tilde{\varphi}(d s(a)) = \varphi(d)$ gives $\delta \tilde{\varphi}(g, h) = -\varphi(\omega(a, b))$, so $tg(\varphi) = -\varphi_*[\omega]$ in your conventions. The general twisted case is *not* “identical” (issue F3).

- Base edge as home of $[\omega]$: **fine once reworted**. $[\omega] \in H^2(\Gamma; D)$ does not live in $E_2^{2,0} = H^2(\Gamma; M^D)$; what lives there is $\text{im}(tg) = \{\varphi_*[\omega]\}$. Say that.
- Fibre edge as home of prunability: **not established, and false on $K_{2,3}$** (fatal issue X1). No map from prunability data into $E_2^{0,2}$ is ever constructed, and on your own example the fibre edge is the *same group with the same page* for the prunable control as for $K_{2,3}$.
- “The direct sum (1) holds iff $[\omega] = 0$ ”: **false in both directions** (issue F1); Theorem 7 (the no-go with the hypothesis $\varphi_*[\omega] \neq 0$) is the correct statement.

Q2. Is “prunability = nonsplitting of $\mathbb{Z}/4$ over $\mathbb{Z}/2$ ” the right skew-brace-side statement? No. Two separate problems.

(a) *It is the wrong level: class versus cocycle.* Let $(A, +)$ be the additive group, $B \leq A$ a subgroup and $L \ni 0$ a loop set that is a transversal of B . Then L defines a normalised section $s_L : A/B \rightarrow A$ with Schreier cocycle $f_L(a, b) = s_L(a) + s_L(b) - s_L(a + b) \in B$, and

$$L \text{ is an additive subgroup} \iff f_L = 0 \text{ as a cocycle.}$$

The class $[f_L] \in H^2(A/B; B)$ is independent of L ; it is the obstruction to *some* transversal being a subgroup (existence of a complement). So

$$\text{nonsplit} \implies \text{no transversal of } B \text{ is prunable,} \quad \text{but not conversely.}$$

Counterexample to the converse: $A = \mathbb{Z}/6$, $B = \{0, 3\}$, $L = \{0, 1, 2\}$. The extension splits ($\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$), yet L is not a subgroup ($1 + 2 = 3 \notin L$); $f_L(1, 2) = f_L(2, 1) = f_L(2, 2) = 3 \neq 0$ (script output). And the class does not see the loop set at all: in $(\mathbb{Z}/4, +)$ the set $\{0, 2\} = B$ is a subgroup (prunable), while $\{0, 1\}$ and $\{0, 3\}$ are not; the $\mathbb{Z}/4$ class is the same nonzero element in every case. So what §7 actually proves is the correct but much weaker statement: *on $K_{2,3}$, both loop sets are transversals of $\{0, 2\}$ in $(\mathbb{Z}/4, +)$, and since $\mathbb{Z}/4$ does not split over $\{0, 2\}$, neither can be a subgroup.* That is a tidy proof of Ferri’s Counterexample 5.2; it is not a cohomological characterisation of prunability.

(b) *It lives in a different extension from the LHS fibre.* The LHS sequence in Theorem 1 is built from the *vertex group* π of the groupoid, i.e. from groupoid composition (the multiplicative/dynamical side). The extension $0 \rightarrow A_2 \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$ is an extension of the *additive group of the brace*. Nothing in the paper maps one to the other; the “identification” in §7 is that both H^2 ’s are abstractly $\mathbb{Z}/2$.

Does the non-group star break it? Not on the additive side: prunability is tested in $(A, +)$, so the quasigroup nature of $\bullet_\lambda$ is irrelevant to statement (a). What the dynamical star does do is remove any reason to expect the additive extension to appear inside the LHS of the vertex group: the groupoid composition is where the star lives, and the star is not the additive group. There was never an identification to break. If you want the additive obstruction as a cohomology class with the right functoriality, its natural home is the *additive* component of a skew-brace H^2 (the β -slot of the (β, τ) pairs you use in your 25 Sep note), with prunability as vanishing of a specific cocycle and “prunable after re-choosing loops” as vanishing of its class. That is a proposal, not a checked claim.

Issues

Fatal

- X1. **The fibre edge cannot detect prunability on $K_{2,3}$ vs. its Klein control** (p. 3, Thm 2; pp. 7–8, §7 “The control”; p. 8, Thm 10(1)).

“A prunable Klein-four control kills both edges. Thus the fibre edge is the genuine detector of prunability” (p. 3); “The control isolates the fibre edge as the genuine detector of prunability: it is nonzero exactly when G fails to be prunable.” (p. 8)

By your own description, in both $K_{2,3}$ and the control the vertex group at S_2 is $\text{loops}(S_2)$, of order 2, so $\pi \cong \mathbb{Z}/2$; by your own §7, $\Gamma = 1$ (so $D = \pi$) and $M = \mathbb{Z}/2$ trivial. Hence the two LHS spectral sequences are *isomorphic*, with $E_2^{0,2} = H^2(\mathbb{Z}/2; \mathbb{F}_2) \cong \mathbb{F}_2 \neq 0$ in *both* cases. The control does not “kill” the fibre edge. More bluntly: both groupoids are equivalent to $B(\mathbb{Z}/2)$, and $H_{cat}^*(-; A)$ with these coefficients is an equivalence invariant, so *no* invariant of $H_{cat}^2(G; A)$ or its LHS page can distinguish them. Prunability depends on how the loops sit inside $(A, +)$, which the vertex-group LHS does not see. The element you call $[\pi]$ is the class of the additive extension of $(\mathbb{Z}/4, +)$, i.e. an element of a different group $H^2((\mathbb{Z}/4)/A_2; A_2)$ that happens also to be $\mathbb{Z}/2$. Theorem 2, the “Answer to Remark 3.20” (p. 8), Theorem 10(1), and the abstract’s “the fibre edge carries a nonzero class $[\pi]$, while a prunable Klein control kills both edges” all fall with this. (The same objection applies to the §1 sentence that prunability “lives in ... $H_{CatB}^2(G) \cong H^2(\pi)$ ” (p. 2), unless the coefficient module is built from the additive structure — which it is not here.)

- X2. **The proposed detector is wrong even on the additive side** (p. 8, Answer to Remark 3.20).

“the prunability obstruction is the fibre-edge class $[\pi] \in H^2(D; M)^\Gamma$..., equivalently the nonsplitting of the additive star extension $0 \rightarrow A_2 \rightarrow (\mathbb{Z}/4, +) \rightarrow \mathbb{Z}/2 \rightarrow 0$.”

See Q2(a): prunability is cocycle-level, the class is extension-level. The class is determined by (A, A_2) alone and is the same for the prunable loop set $\{0, 2\} \subset \mathbb{Z}/4$; the converse direction fails for $\mathbb{Z}/6 \supset \{0, 3\}$, $L = \{0, 1, 2\}$. And the Klein control is confounded: it changes the additive group $(\mathbb{Z}/4 \rightarrow V_4)$, not just prunability. A valid control would keep $(\mathbb{Z}/4, +)$ and change the loop sets.

Fixable

- F1. **“(1) holds iff $[\omega] = 0$ ” is false both ways** (p. 2, Thm 1; p. 8, Thm 10 last sentence).

“Consequently the direct sum (1) holds if and only if $[\omega] = 0$ ”; “The clean direct sum (1) holds iff this coupling vanishes, i.e. iff $[\omega] = 0$.”

($\Leftarrow$ fails): $\pi = \mathbb{Z}/2 \times \mathbb{Z}/2$, $M = \mathbb{F}_2$: $[\omega] = 0$ but $\dim H^2 = 3 > 1 + 1 = \dim E_2^{2,0} + \dim E_2^{0,2}$; the cross term $E_2^{1,1}$ is nonzero, as your own Corollary 8 table records (“1+1+1 = 3 ... clean (full Künneth)”). ($\Rightarrow$ fails): $\pi = \mathbb{Z}/9$, $D = \Gamma = \mathbb{Z}/3$, $M = \mathbb{F}_2$: $[\omega] \neq 0$ but every term is 0, so any “sum” holds vacuously. Replace by Theorem 7’s hypothesis ($\exists \varphi$ with $\varphi_*[\omega] \neq 0$) and decide whether (1) includes $E^{1,1}$. Relatedly, (1) as displayed on p. 2 is $H^2(\pi; M) \oplus H_{base}^2$ with $H^2(\pi; M)$ as the prunability summand — but $H_{cat}^2 = H^2(\pi; M)$ by Prop. 3(b), so (1) is ill-posed as written; Theorem 1(ii) instead puts prunability in $H^2(D; M)^\Gamma$. Pick one.

F2. **“Connected, so $[\omega] = 0$ ” is a non sequitur** (p. 3, Thm 2; p. 7, “Base edge”).

“ $[\omega] = 0$ (G is connected, so the orbit base is contractible)”; “ $K_{2,3}$ is connected, so the orbit base is the coarse groupoid on $\{S_2, S_3\}$ ”.

Theorem 1 *assumes* G connected; if connectedness forced $\Gamma = 1$ the base edge would vanish for every groupoid in the theorem. What you need is $D = \pi$ (full loop bundle) for the Zappa–Szép factorisation of $K_{2,3}$; state what that factorisation is and prove $D = \pi$. And note the consequence: with $\Gamma = 1$ the spectral sequence is degenerate ($E_2^{0,2} = H^2(\pi; M)$ is the abutment), so the “two edges decouple” on $K_{2,3}$ has no content.

F3. **Lemma 5 in the twisted case** (p. 5).

“the general Γ -module case is [8, transgression formula], identical after keeping the covariant end.”

$H^1(D; M)^\Gamma = \text{Hom}_\Gamma(D, M)$ and $\varphi(D) \subseteq M^D$ require D to act trivially on M ; otherwise $H^1(D; M)$ is crossed homomorphisms modulo principal ones and the formula changes. Put “ D acts trivially on M ” into the statement of Lemma 5 and Theorem 1(iii). Fix the sign ($tg = -\varphi_*[\omega]$ with your cochain conventions) or state the convention.

F4. **The coupling is not a single differential** (p. 9, Remark 11: “orthogonality is replaced by a single coupling differential”). For $\mathbb{Z}/4$ over $\mathbb{Z}/2$ with $\mathbb{F}_2$ coefficients the E_2 total is 3 and H^2 is 1; tg kills one dimension, so $d_2^{1,1} : E_2^{1,1} \rightarrow E_2^{3,0}$ (cup with the extension class) must kill another. Your table’s “naive sum overcounts by 2” (p. 7) already says this. Rephrase: $[\omega]$ drives d_2 on the whole $q = 1$ row, not only the transgression.

F5. **Corollary 6 contradicts itself** (p. 6). The hypothesis is “collapses at E_2 in total degree 2 with vanishing cross term $E_2^{1,1}$ ”, but item (ii) concludes “including the cross term $H^1(\Gamma) \otimes H^1(D)$ that (1) omits”. Either (ii) is not an instance of the corollary or the hypothesis is wrong.

Cosmetic

C1. **Q8 is mislabelled** (p. 7, Cor. 8 table): “ $Q_8 = (\mathbb{Z}/2)^2 \cdot \mathbb{Z}/2$ ”. In your $D \cdot \Gamma$ convention this says $D = (\mathbb{Z}/2)^2$, but Q_8 has a unique involution (checked) and no Klein subgroup. Your numbers $3 + 2 + 1 = 6$ are those of $D = Z(Q_8) = \mathbb{Z}/2$, $\Gamma = (\mathbb{Z}/2)^2$; relabel $Q_8 = \mathbb{Z}/2 \cdot (\mathbb{Z}/2)^2$. The other rows and all $\dim H^2(\pi; \mathbb{F}_2)$ values $(1, 2, 3, 1)$ check out by bar complex.

C2. **Contradictory description of the coarse base** (p. 4): “ $Sk_C = I(\text{ob } G)$ is the coarse groupoid on the objects, a disjoint union of contractible components” — the coarse (indiscrete) groupoid is connected.

C3. **“dim” with non-field coefficients** (p. 6, Thm 7(b)): fine for $M = \mathbb{F}_2$; for general M use order or length.

What would make it ready for publishable-result

1. **Split the paper.** The part that is proved — LHS for the normal Zappa–Szép regime, $\text{im}(tg) = \{\varphi_*[\omega]\}$ for D acting trivially, and the no-go Theorem 7 with the Q8 label fixed — is a clean short note once F1, F3–F5 are fixed. It should say nothing about prunability.

2. **Withdraw** Theorem 2, the “Answer to Remark 3.20”, and Theorem 10(1) in their current form, and do not send them to Ferri or Neil as an answer.
3. **Replace** them with the honest additive statement Q2(a): for loop sets that are transversals of a subgroup $B \leq (A, +)$, $\text{prunability} \iff \text{the Schreier cocycle } f_L \text{ vanishes}$; the class $[f_L] \in H^2(A/B; B)$ obstructs existence of a prunable re-choice; nonsplitting of $\mathbb{Z}/4$ over $\{0, 2\}$ therefore reproves Counterexample 5.2. If you want a cohomological invariant of prunability, it must use coefficients or a complex that sees the additive structure (e.g. the additive component of a skew-brace H^2), not the vertex-group LHS.
4. **Controls:** vary the loop sets inside a fixed $(A, +)$; never vary the additive group and call it a prunability control.

The open node `prunability-is-fibre-edge-class` should be closed as *refuted as stated* (vertex-group fibre edge), not left open.